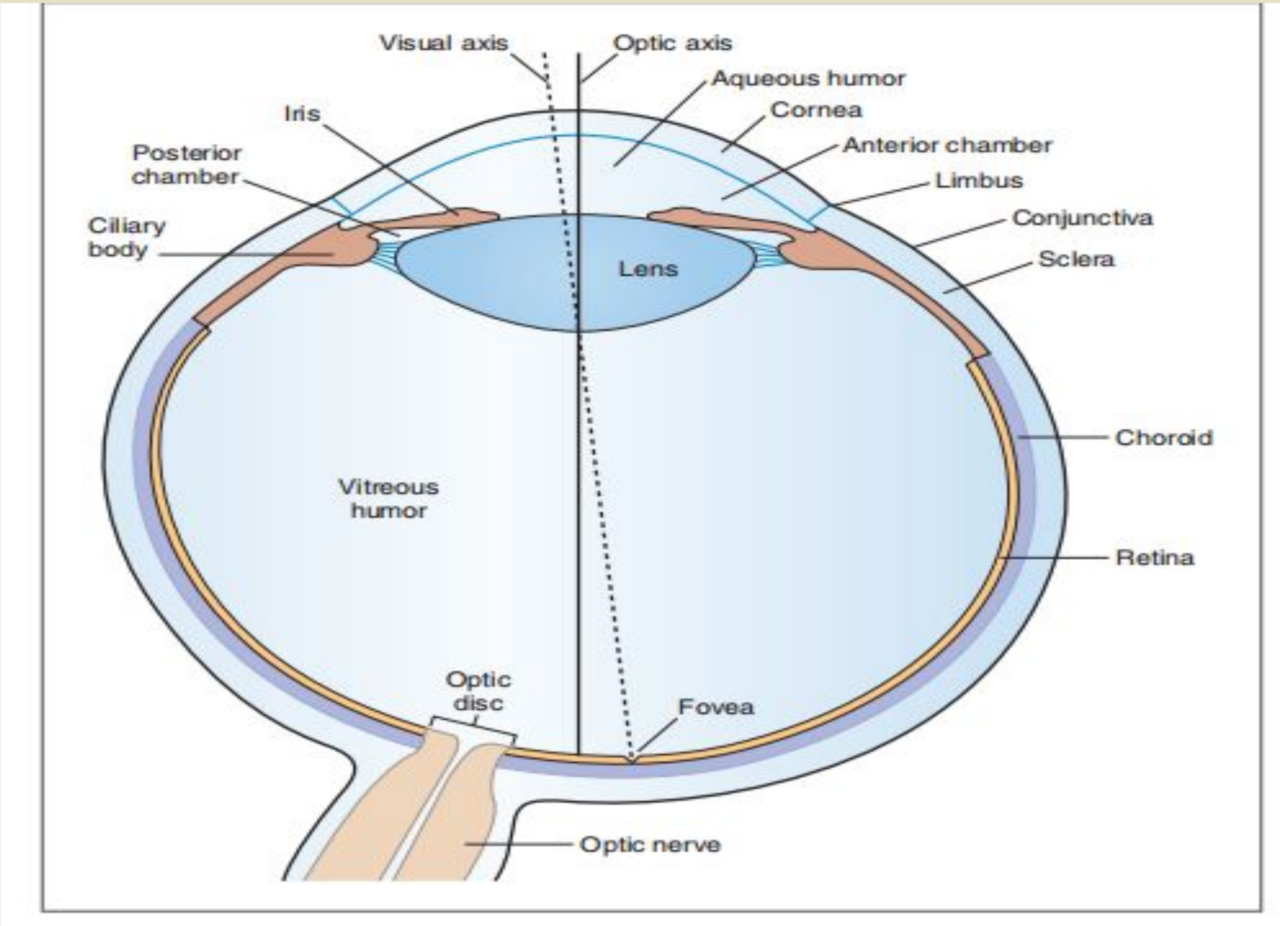
PHYSIOLOGY OF VISION

Learning outcomes

At the end of lecture, a student should be able to:

- Outline the components involved in image formation and explain its mechanism
- Interpret the process of visual accommodation
- Discriminate between errors of refraction and state their corrective mechanisms
- Discuss the formation and functions of aqueous humor
- Illustrate the drainage pathway of aqueous humor and its disorders
- Analyze the functions of photoreceptors and other retinal cells in the retina
- Explain the mechanism of phototransduction, visual pathway and processing

STRUCTURE OF EYE



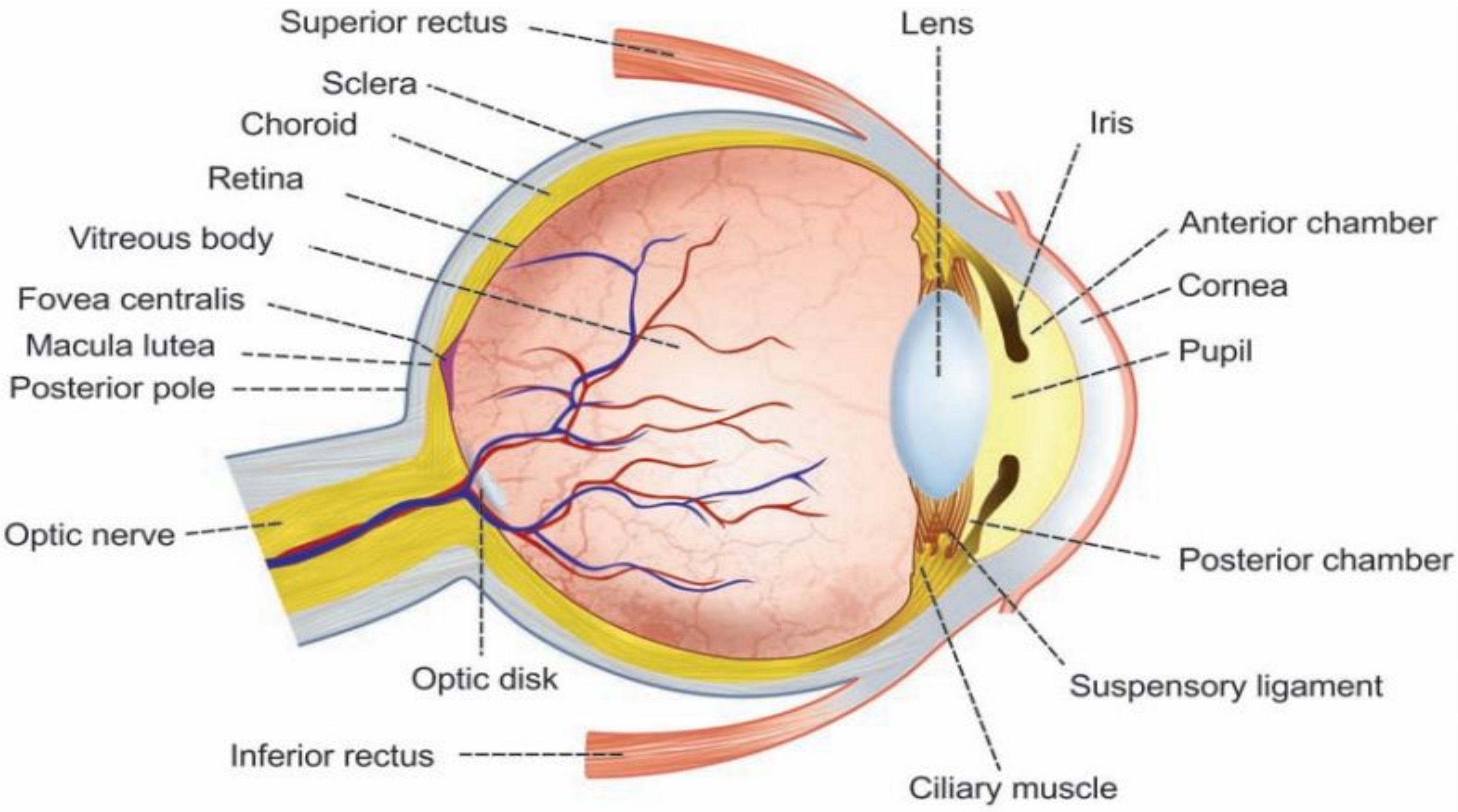


FIGURE 165.2: Structure of eyeball

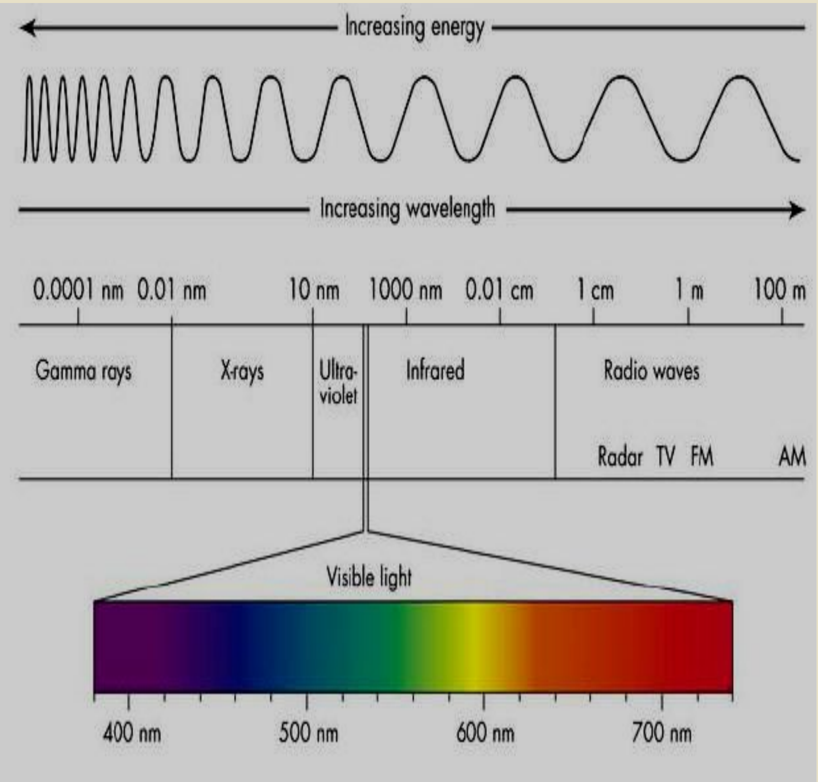
MAJOR STRUCTURES OF EYE

- The wall of the eye consists of three concentric layers: an outer layer, a middle layer, and an inner layer.
- The **outer fibrous layer** includes the cornea, corneal epithelium, conjunctiva, and sclera.
- The **middle vascular layer,** includes the iris and the choroid.
- . The **inner neural layer,** contains the retina. The functional portions of the retina cover the entire posterior eye, with the exception of the blind spot, which is the optic disc (head of the optic nerve).
- Visual acuity is highest at a central point of the retina, called the macula; light is focused at a depression in the macula, called the fovea.
- The eye also contains a lens, which focuses light; pigments, which absorb light and reduce scatter; and two fluids, aqueous and vitreous humors. Aqueous humor fills the anterior chamber of the eye, and vitreous humor fills the posterior chamber of the eye.
- The sensory receptors for vision are photoreceptors,which are located on the retina. There are two types of
- Photoreceptors, rods and cones.Rods have low thresholds, are sensitive to low-intensity light, and function well in darkness. Cones have a higher threshold for light than the rods, operate best in daylight.

Visible spectrum

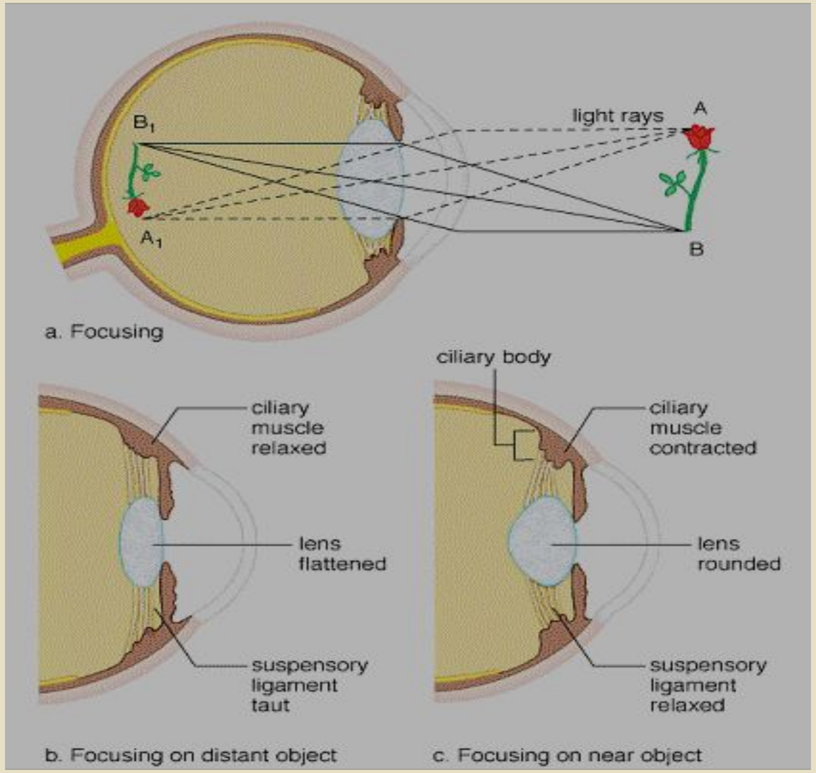
- When light rays strike eyes, the electromagnetic energy is transduced into nerve impulses by photoreceptors
- Each eye has 2 types of receptors: Rods for dim light and cones for bright light and color vision

- The eye can distinguish 2 qualities of light: brightness & wavelength.
  For human eye, the wavelengths between 400 and 750nm are called visible light. Eye is unable to perceive infra red or ultraviolet rays.



Mechanism of image formation

- To form a clear image on retina, 3 mechanisms are important
- 1-Refraction or bending of light rays at anterior and posterior surfaces of cornea and both surfaces of lens
- 2-Constriction of pupil (occurs with refraction and accommodation)
- 3-Accomodation of lens and medial convergence of eyeballs



1-Refraction

- Light rays usually bend when they enter from one transparent medium to another with different density
- The ratio of light velocity in a vacuum to velocity in a different medium is known as refractive index (RI)
- Refractive index of Air: 1.00

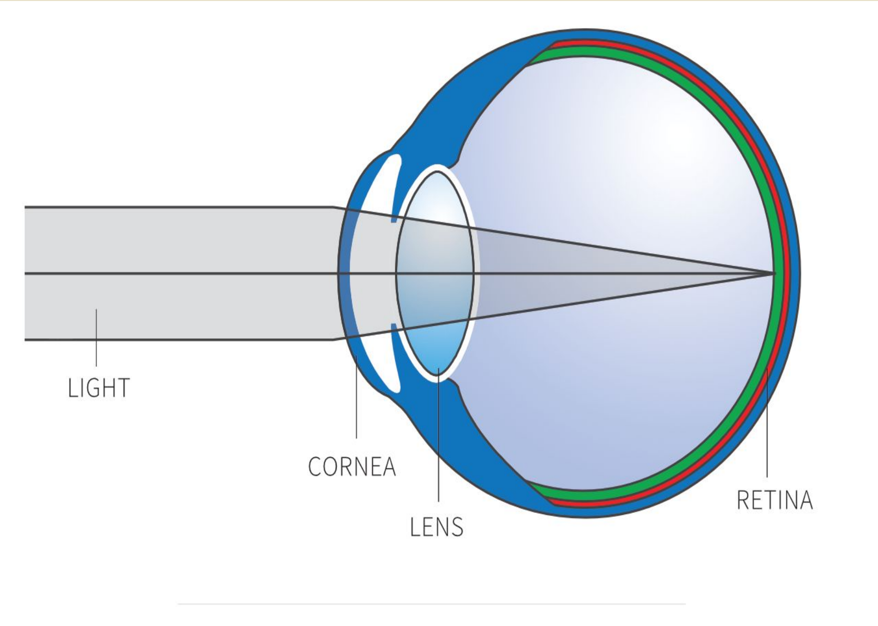
Water: 1.33

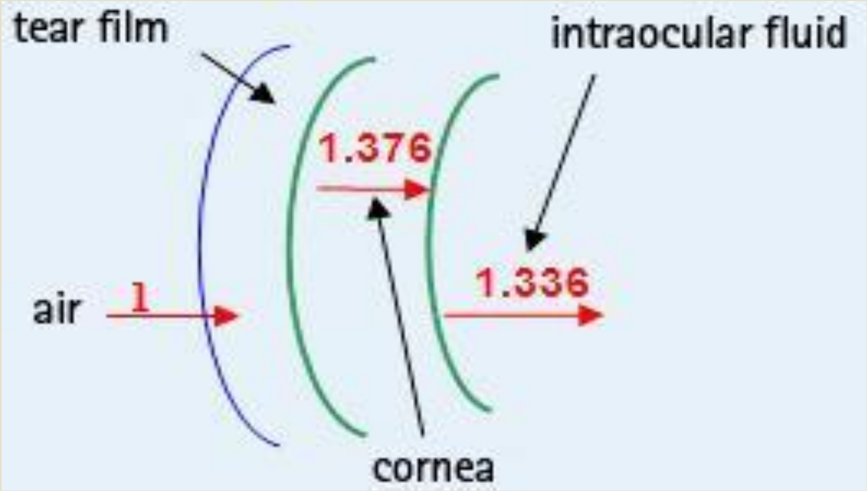
Cornea: 1.38

Aqueous & Vitreous humor 1.33

Crystalline lens: 1.40

(RI of lens and cornea are closer to 1.4)





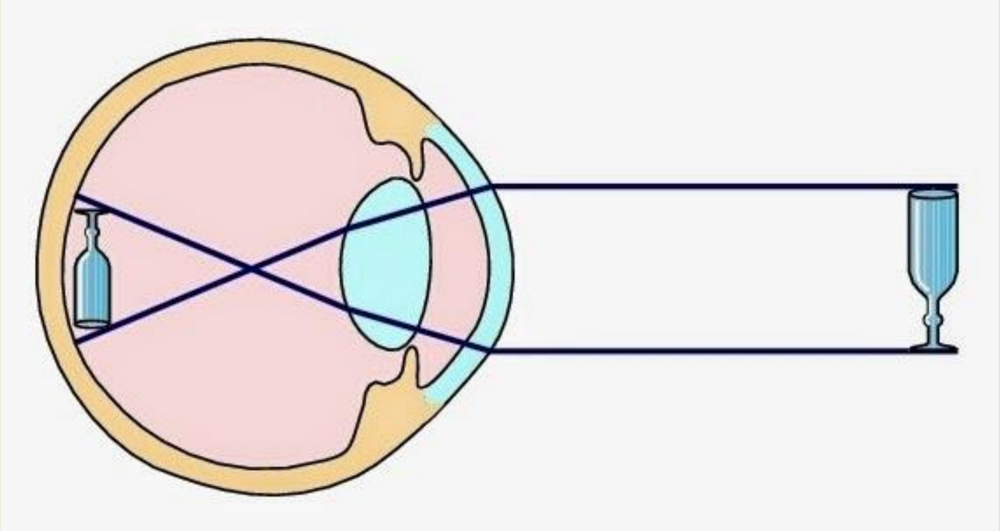
Factors affecting refraction

- Refraction of light brings objects into exact focus on retina. Curvature of cornea remains same but that of lens changes during focusing
- 2 factors contribute to refraction

1. Comparative densities of two media; greater the difference in density, greater is the degree of bending.

2. Angle at which light strikes; Greater the angle, greater will be refraction.

- 75% of total refraction occurs at cornea because of greatest difference in RI at the air-corneal junction (1.00:1.38)

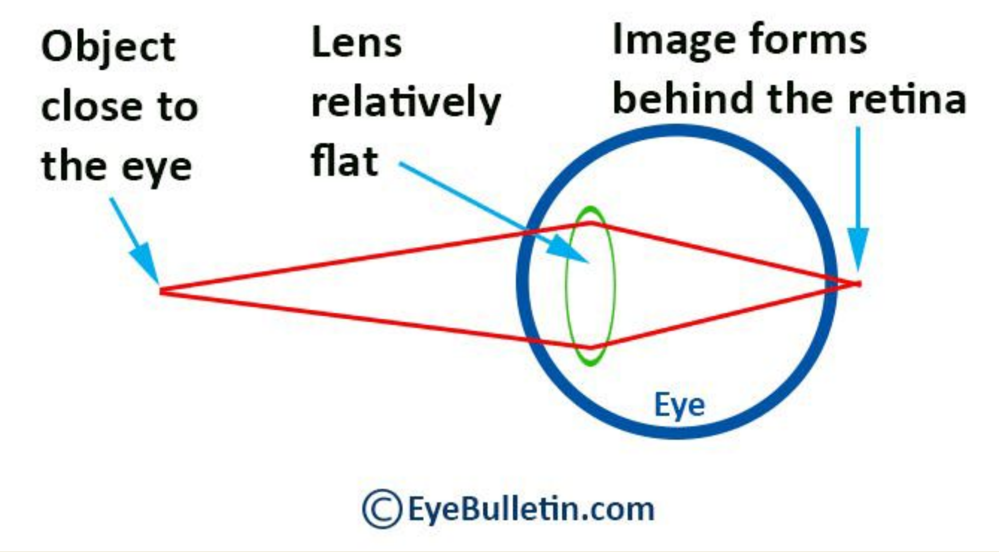


Lens

- Lens is transparent because

1. It is avascular
2. Cell organelles are destroyed
3. Cytoplasm is filled with crystalline protein

- Lens is biconvex, refracts rays towards each other, intersect at focal point; then focus them on retina.
- Anterior surface curvature:11-12 mm for far vision and 6-7 mm during accommodation
- Posterior surface radius is 6 mm at rest and 5.5 mm in full accommodation

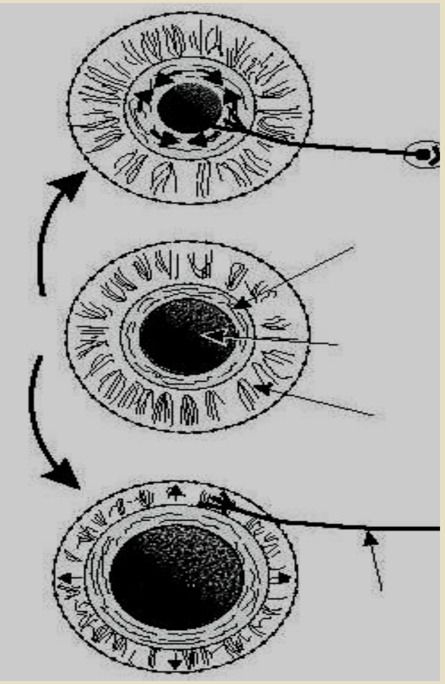


2- Constriction of pupil

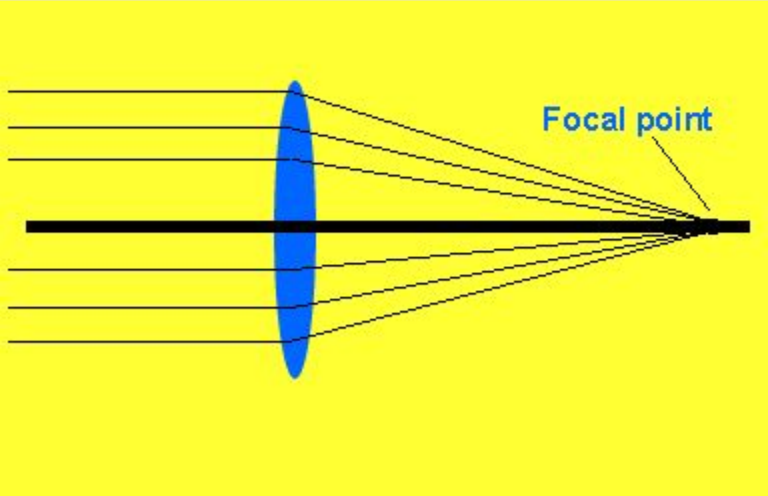
- Major function of iris is to ↑ or ↓ amount of light entering the eye by constriction or dilation of pupil
  1. Quantity of light changes 30 folds &
  2. Gaze shifts b/w distant & near vision

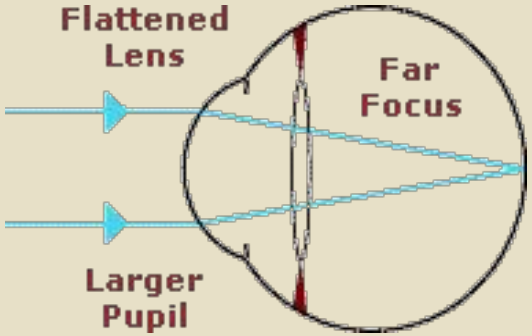
Pupillary reflex: Circular muscles (parasympathetic) contract to decrease pupil diameter (meiosis); radial muscles (sympathetic) contract to dilate pupil (mydriasis)

- Range of pupil diameter : 1.5 to 8 mm
- Pupillary constriction increases depth of focus, because all light rays pass through center of the lens.



3- Accommodation





- Is the ability of eye to keep image focused on retina as the distance between eye and the object varies
- The curvature of lens changes to enable the eye to keep object focused. Greater the curvature of lens greater the degree of bending and stronger the lens: thus, Lens fine tunes the image.

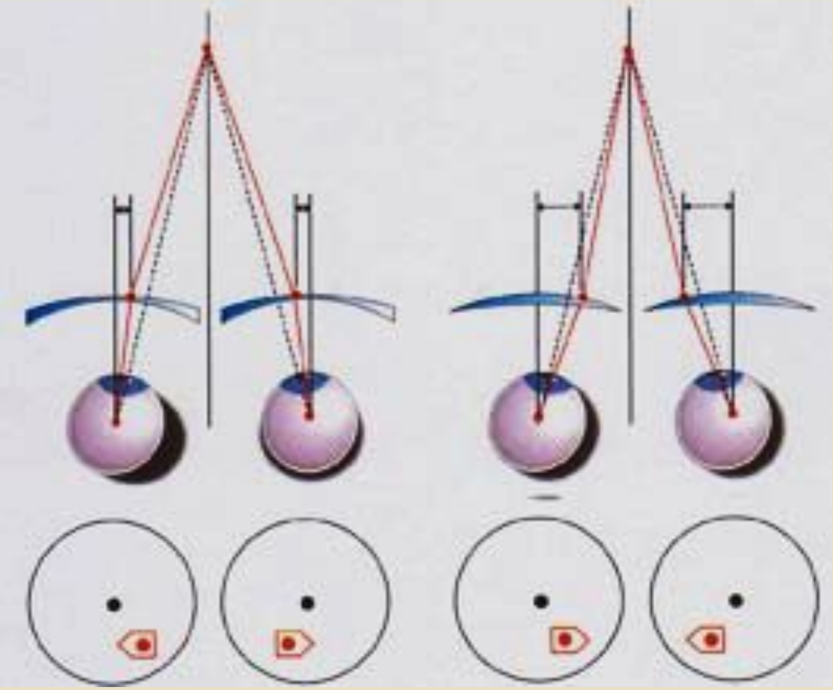
- Mechanism of accommodation
- Far vision: Ciliary muscles are relaxed, lens is flat by taut suspensory ligaments, the object is focused on the retina far from the lens
- Near vision: Ciliary muscles contract, suspensory ligaments relax; as lens is elastic it becomes more convex, increasing its curvature and focusing power. The object is focused on the retina by greater convergence of light rays

Convergence of eyes for near vision

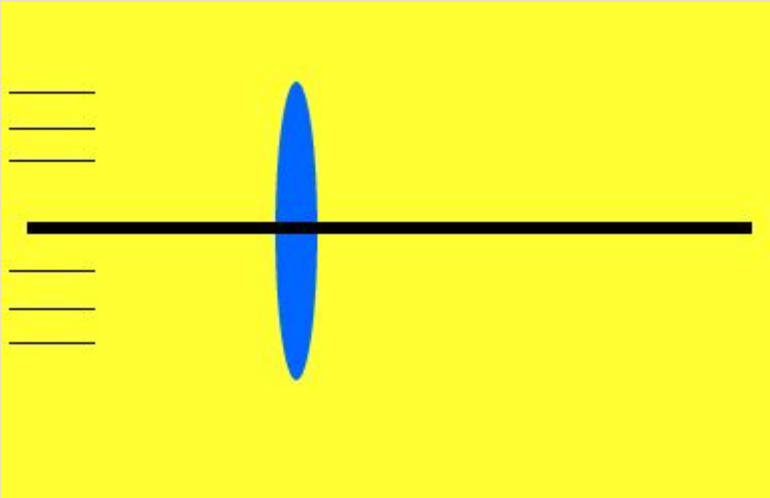
- When eyes are focused on near objects, both eyeballs move medially due to simultaneous contraction of medial recti and eyes rotate medially

- Visual axis of each eye moves towards the object to focus the image on each fovea

- If eyes can not converge accurately double vision or diplopia results; images fall on different parts of two retinae and brain sees 2 images



Refractive [focal] power of eye (cornea & lens)



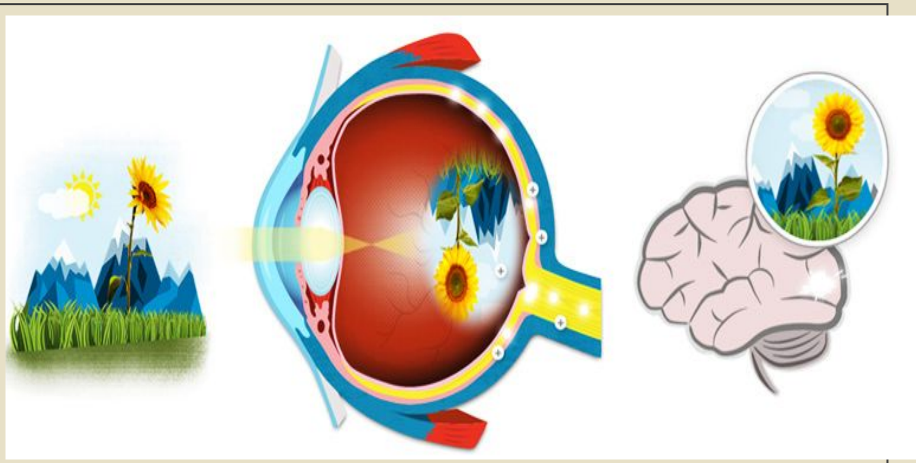
- The distance from center of lens to principal focus is known as focal length of lens. It varies with thickness of lens.
- Focal power is the inverse of focal length and measured in Diopters (D)
- A diopter is $= \dfrac{1}{\text{Focal length of lens in meter}} = \dfrac{1}{17/1000} =$ **59 D**

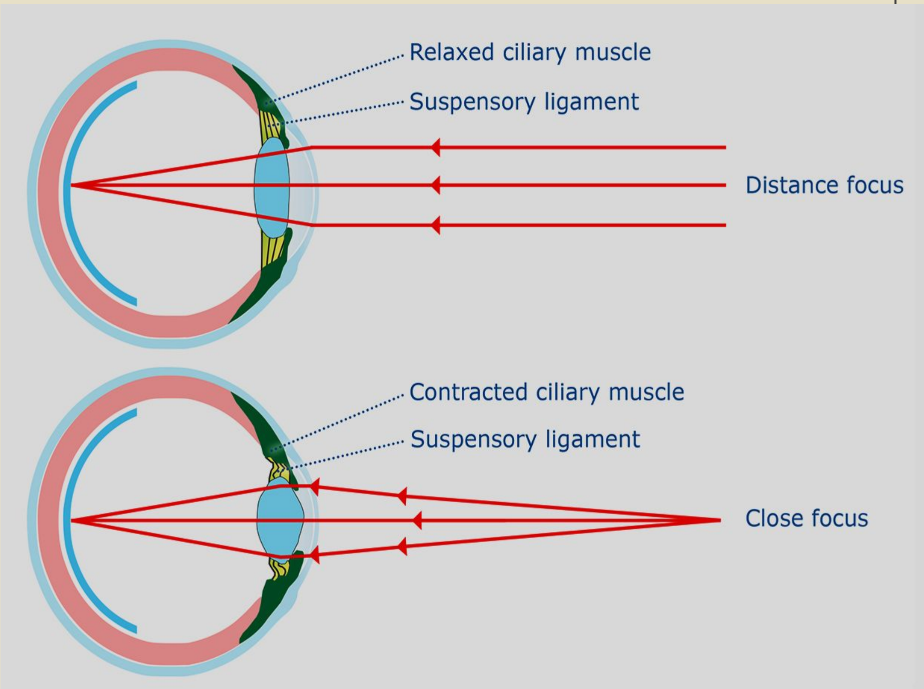
Cornea = 43 D; lens = 16 D [far vision]

- In maximum accommodation for near vision in young person with normal vision, refractive power increases by 10-14 D

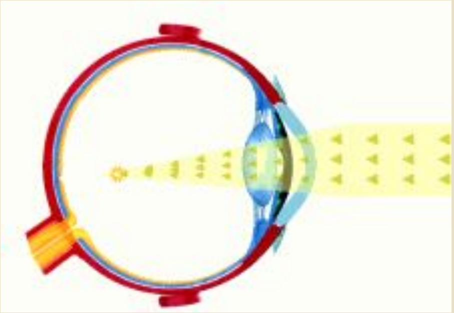
Image formation

- **Image**: is real, inverted and undergoes right to left reversal i.e., light from right side of object strikes left side of retina and left side of object strikes right side of retina
- Visual acuity: ability to distinguish 2 points as separate
- Near point: minimum distance to which eye can accommodate; for a young person 10 cm
- Far point: when object placed 6 meters away, is seen clearly without accommodation. Normal vision: 6/6 m
- If image is focused behind or in front of retina, it is blurred; a condition called Error of refraction





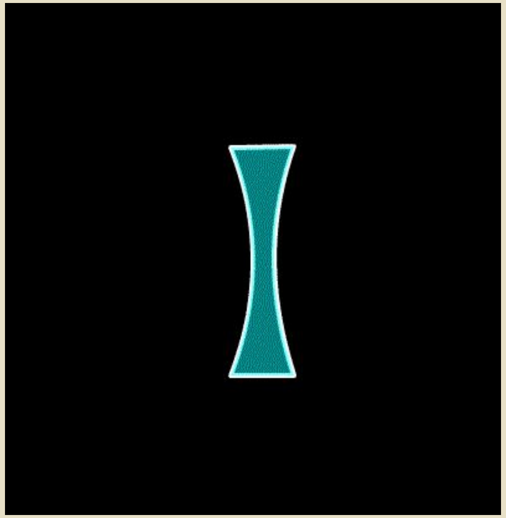
1. MYOPIA [short sightedness]



- For far vision parallel light rays from a distant point pass through nodal point, focused on retina.
- In this condition image is focused in front of retina and blurred.
- Better near vision than far vision

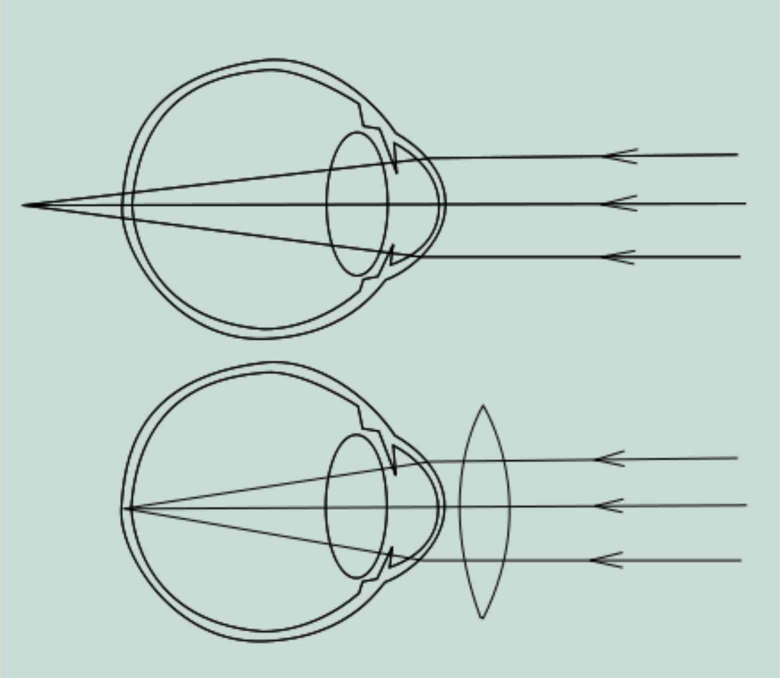
Causes:

1. Eyeball is long antero-posteriorly
2. Lens is too strong



Corrected by biconcave lens with negative diopter

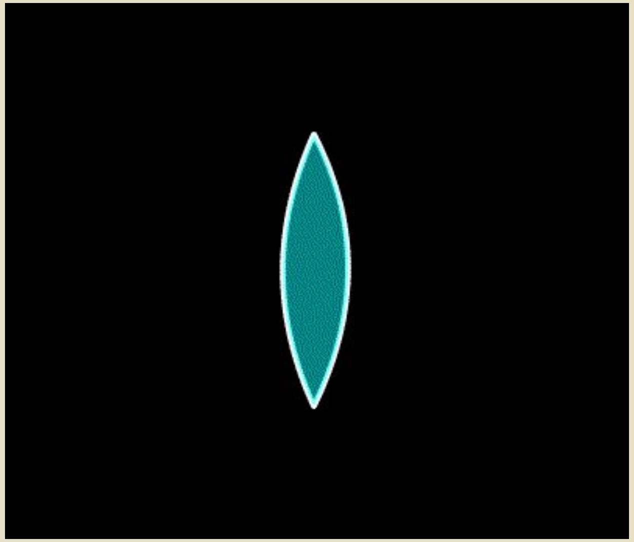
2. HYPERMETROPIA [far sightedness]



- Near objects are focused behind the retina even with accommodation and are blurred

Causes:

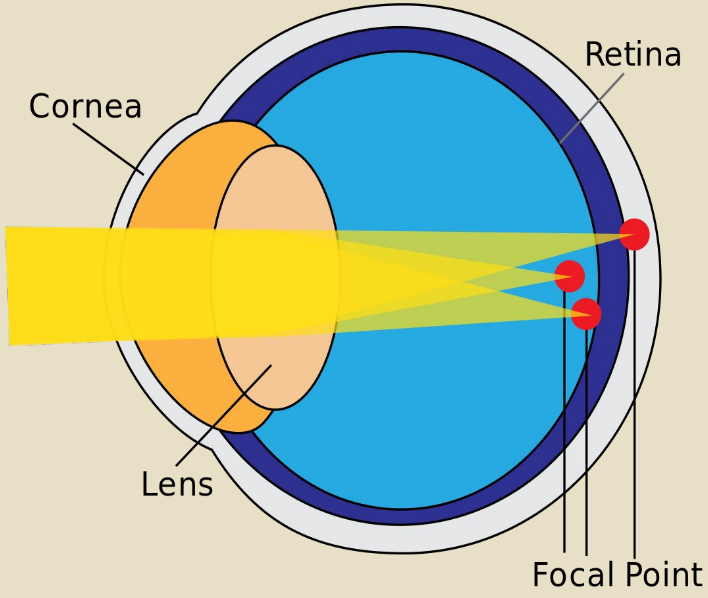
- Eyeball is too short or
- lens is too weak
- Better far vision than near vision



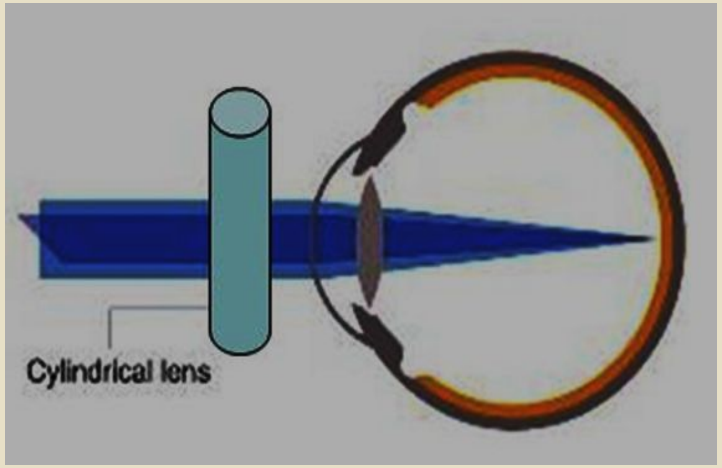
Corrected by using biconvex lens [+ diopters]

3. ASTIGMATISM

- Curvature of cornea is uneven, more elongated in one direction
- Light rays unequally refracted resulting in blurring of vision as some light rays are focused in front and some behind the retina.

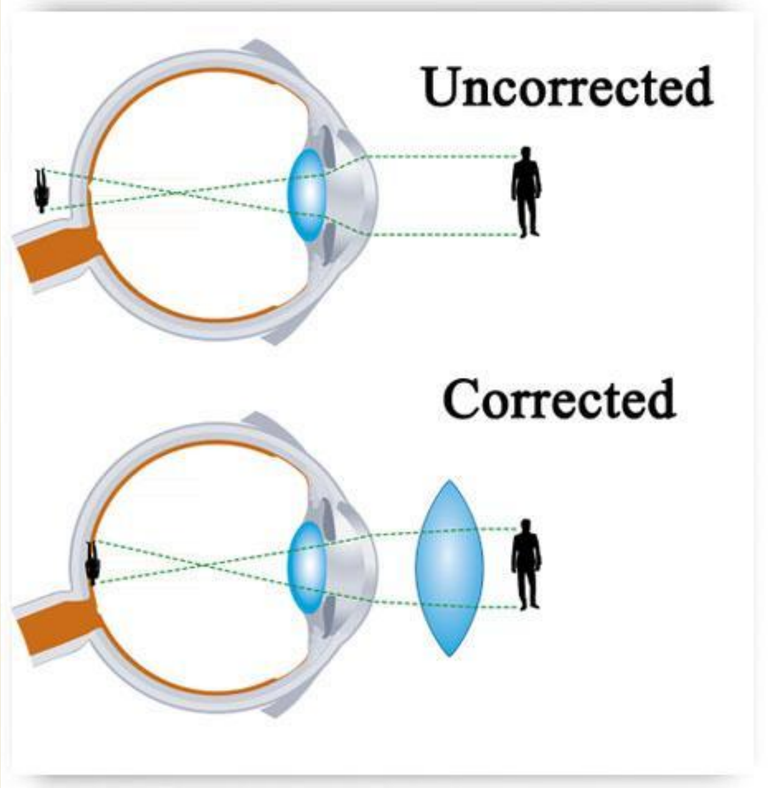


- Regular astigmatism is corrected by **cylindrical lens** which has more curvature in one direction (converging power in one axis only)
- Irregular astigmatism caused by scars can be corrected by **contact lenses**



4. PRESBYOPIA

- With aging, lens loses its elasticity, fails to increase power and its ability to accommodate
- Presbyopia begins at the age of mid 40s and people require reading glasses
- At 40 years age near point may reach 20 cm and at age 60 up to 80 cm
- **Corrected by using biconvex lens**

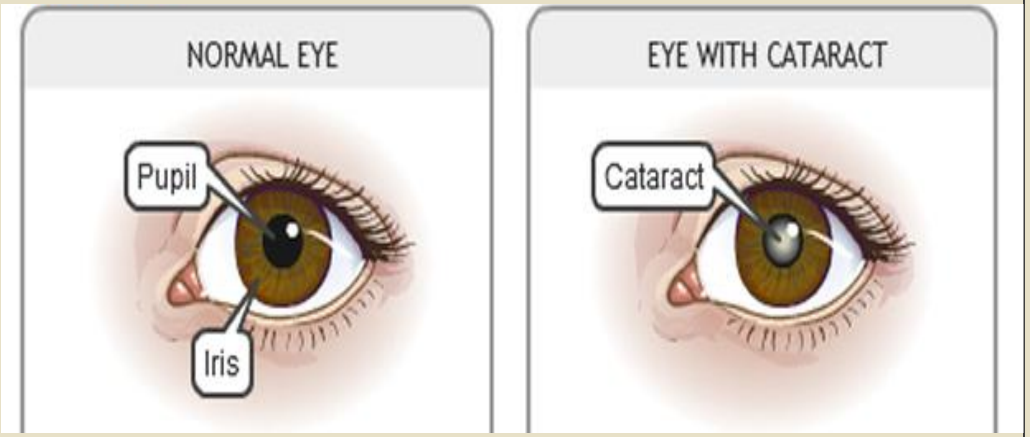


Advantages of Contact lenses

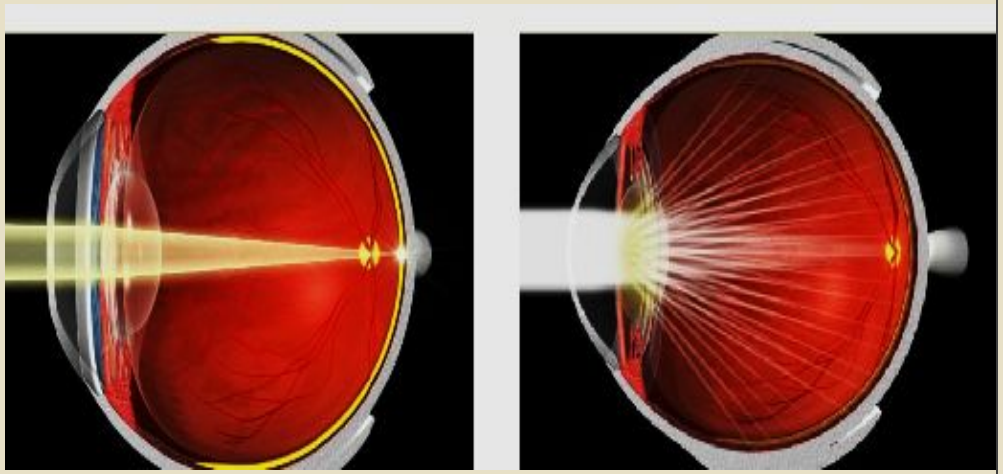
- Held in place by a thin layer of tear fluid.
- Outer surface of contact lens plays a major role in refraction
- Lens turns with eyes, thus provides a broader field of clear vision than glasses
- Contact lens has little effect on size of object whereas a glass lens placed 1 cm or so in front of eye affect size of image
- Suitable for patients with odd shaped or bulging cornea (keratoconus) as no glasses can correct vision satisfactorily

Cataract

- With advancing age, there is denaturation of lens proteins, lens becomes inelastic, opaque and cloudy, so that light rays can not pass through it, a condition called cataract
- Vision appear milky or as if one were looking from behind a waterfall



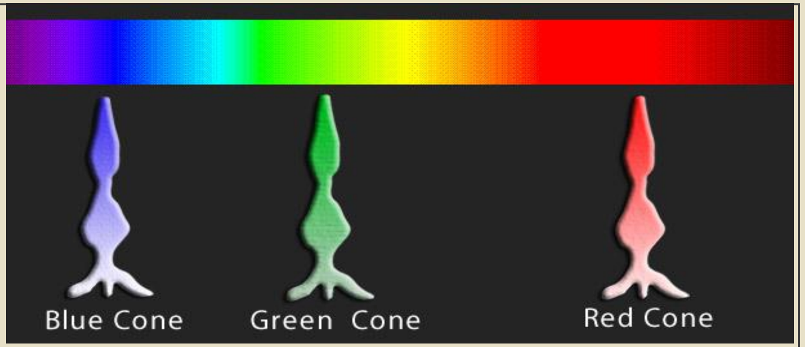
- The cataract may be congenital, or due to metabolic disorders or trauma.
- Opaque lens is surgically removed and replaced by intraocular lens or convex lens [glasses] to correct the vision

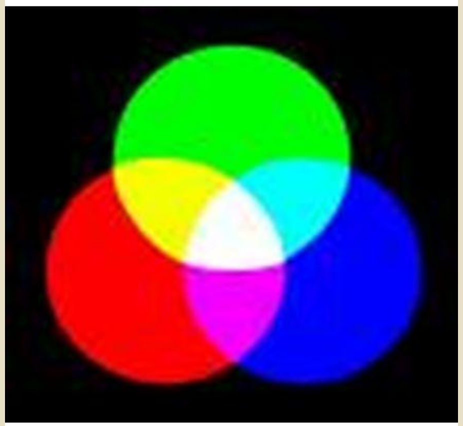


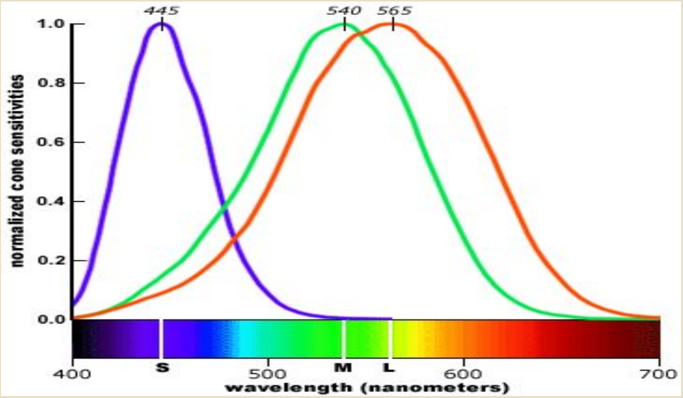
Color vision

- Cones are responsible for color vision. Each cone is sensitive to one of the 3 primary colors (red, blue, green)
- When cones are stimulated in combinations, the brain perceive more than 200 different colors; e.g. for yellow color, both red and green cones are stimulated equally

- Red color has longest wavelength and blue cones have shortest wavelength
- Color blindness is an X-linked disease ;thus more common in males. Green-red blindness is common







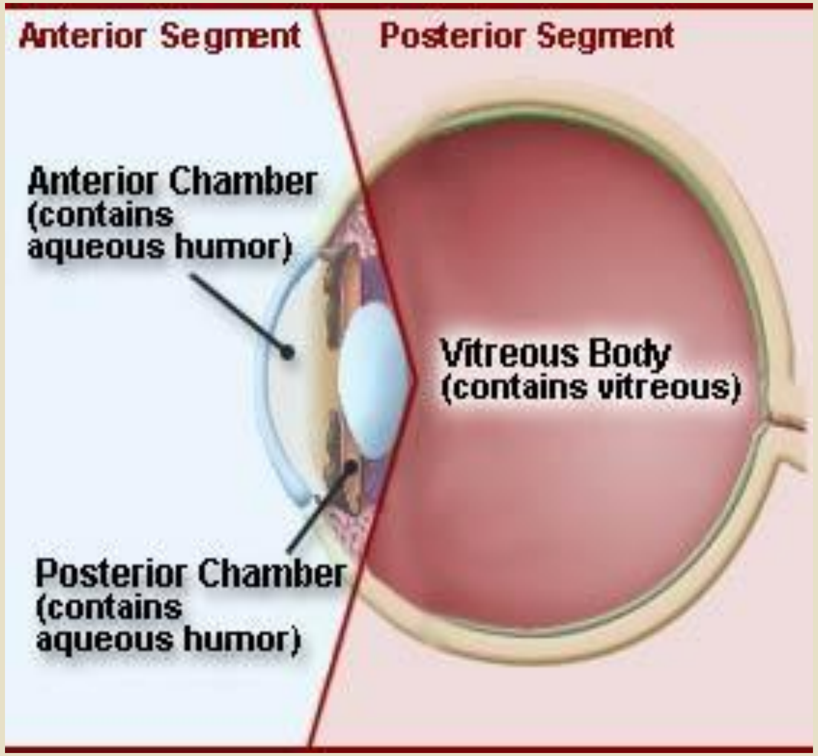
Intraocular fluid

Aqueous humor: Watery fluid, located between cornea and lens in anterior and posterior chambers [anterior cavity]

- Provides nutrition to lens, cornea and iris
- Removes metabolically toxic products
- Gives refractive index of 1.33
- Maintains intraocular pressure (IOP)
- Ascorbate-anti-oxidant, provides UV protection
- Facilitates cellular and humoral response of eye to infection and inflammation

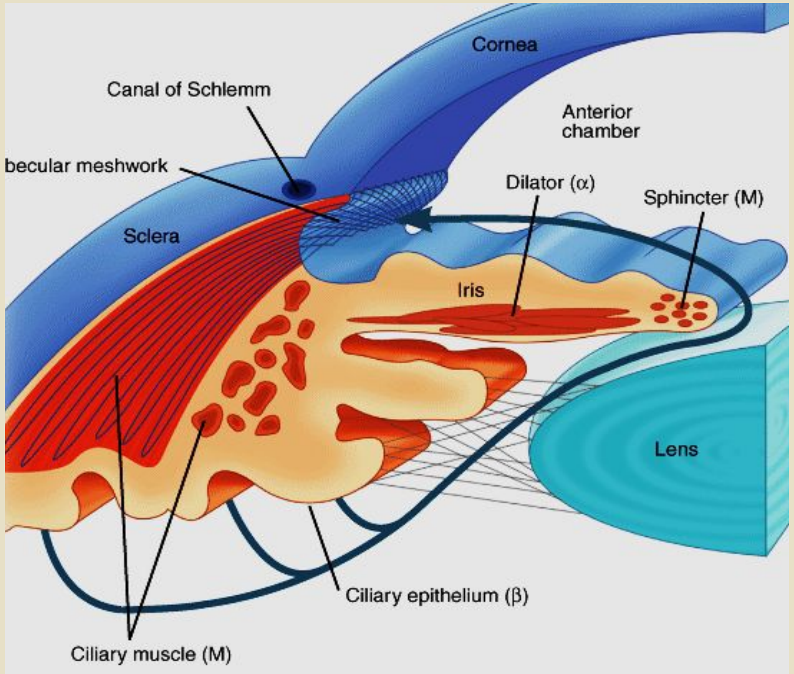
Vitreous humor: Transparent gelatinous substance composed of water & proteins that fills the posterior cavity between lens and retina. Does not circulate.

Useful forensically if other body fluids not available



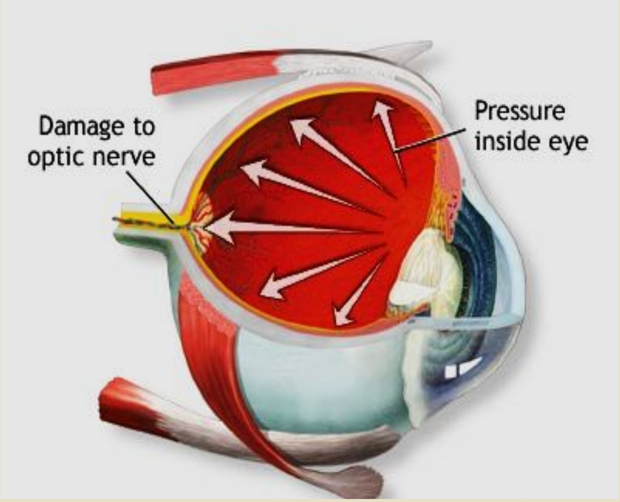
Formation and circulation of aqueous humor

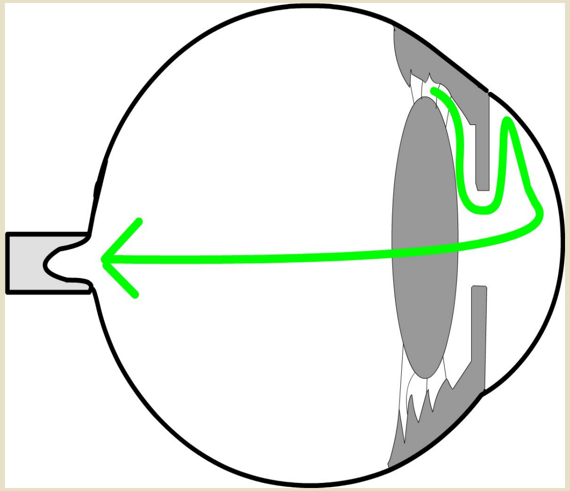
- Produced by active secretion of epithelial cell of vascularized ciliary process in posterior chamber.
- Active transport of Na^+ pulls chloride and HCO3 ions and these ions cause osmosis of water from blood capillaries
- AH flows through pupil into anterior chamber, anterior to lens, into angle b/w cornea and iris, then through trabecular meshwork into canal of Schlemm which empties into extraocular veins.
- The drainage of aqueous humor must balance its production to maintain a constant ocular pressure
- Normal I/O P is 15 ± 2 mmHg (range: 12-20 mmHg).
- It is measured by Tonometer



Glaucoma

- IOP greater than 25-30: causes compression & ischemia of optic nerve with progressive vision loss
- Dangerously high pressure 60-70 mmHg causes blindness
- Glaucoma: is an optic neuropathy, one of the most common causes of blindness
- During infection or hemorrhage, debris in AH accumulates in trabecular spaces and prevent adequate reabsorption of fluid causing glaucoma: removed by phagocytes
- Outflow obstruction due to any cause leads to increased I/O pressure, compress and damage axons in optic nerve at optic disc irreversibly. Compression of retinal artery damages neurons by depriving them from nutrition



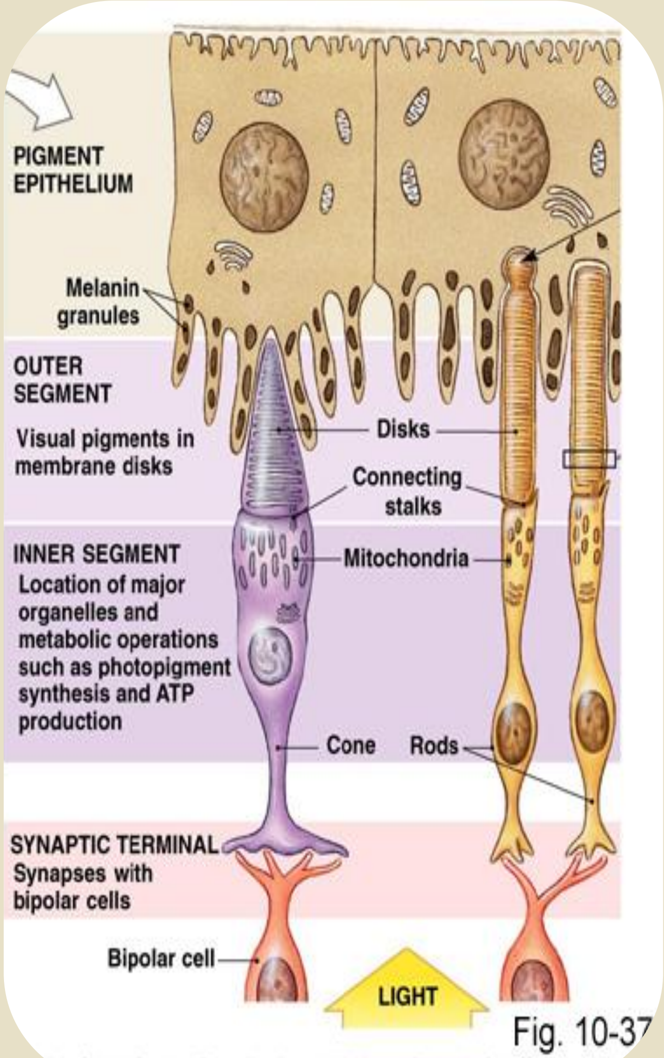


Photoreceptors

- Each eye has 130 million rods for Scotopic/ night vision, 5.5 million cones for Photopic/bright light and color vision
- PR contains photopigment that absorbs light and becomes active
- It has 2 components:
- Opsin: integral protein of disc membrane
- Retinal: vitamin A derivative binds interior of opsin (light absorbing part
- Rods: Rhodopsin (retinene + scotopsin)

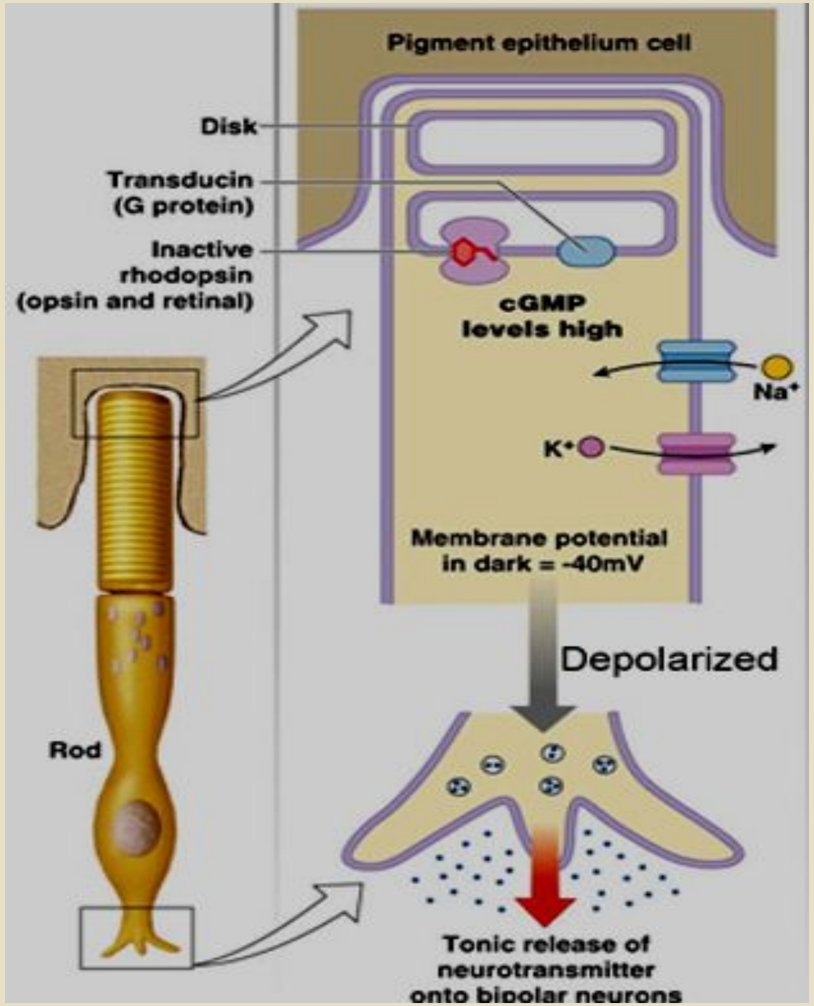
  Opsin in 3 cones called Photopsin 1. Propyropsin 2. Iodopsin 3. Cyanopsin

- Red cones respond to longest & blue cones to shortest wavelengths of light



Photoreceptors are depolarized during dark

- Plasma membrane of discs contains ligand gated Na^+ channels that bind an internal 2^{nd} messenger cGMP which allows Na^+ influx in dark
- This Na (dark) current reduces membrane potential from -70 to -40 mv
- Depolarization spreads to synaptic terminal, open Ca^{++} channel
- Inhibitory neurotransmitter Glutamate is secreted which inhibits bipolar cells
- No action potential during dark



Receptors are hyperpolarized by light

- Light on retina converts 11cis-retinal to all-trans retinal a process called photoisomerization. A number of intermediates are formed, one of them is metarhodopsin II.
- It activates Transducin, a G protein which in turn activates intracellular enzyme Phosphodiesterase which degrades cGMP into 5′ GMP
- Reduction in cGMP closes Na channels and hyperpolarize receptor membrane to -70 mv
- Hyperpolarization decreases glutamate release and disinhibit bipolar cells which are stimulated

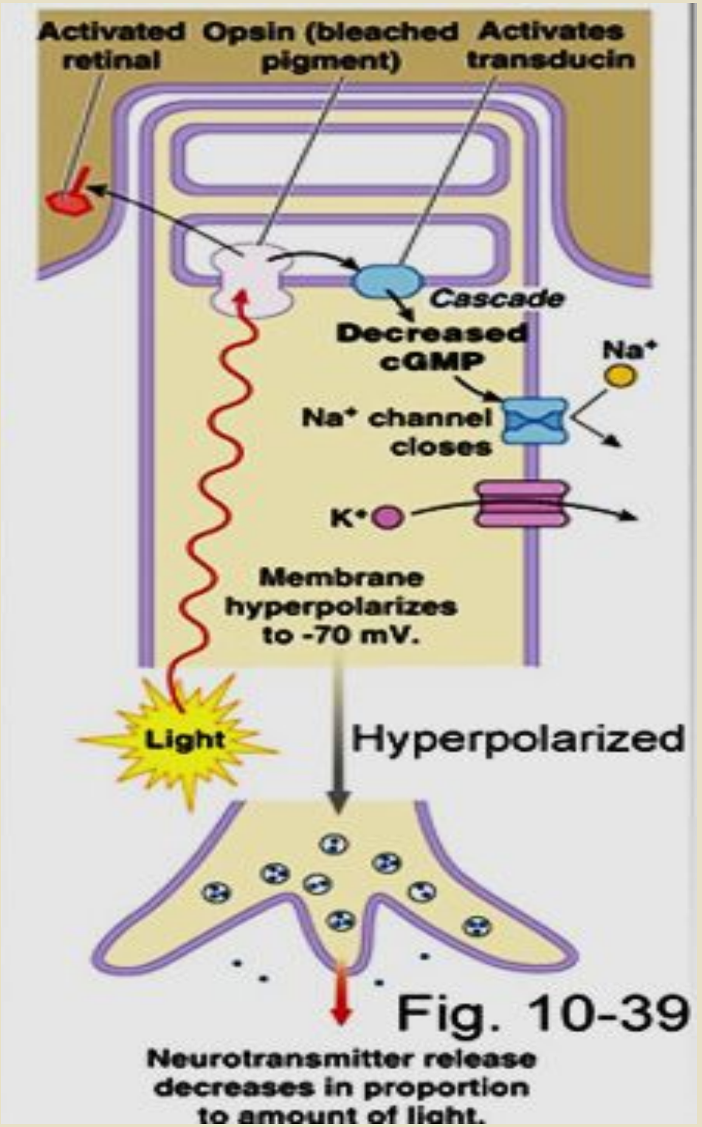
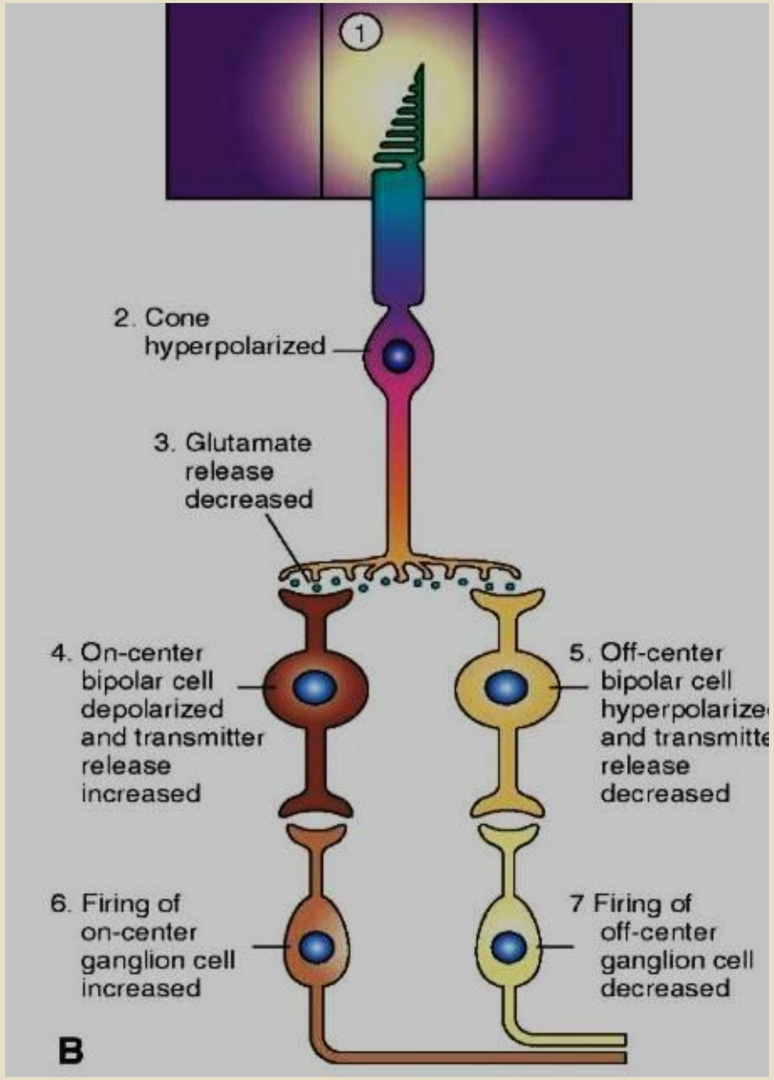


Fig. 10-39

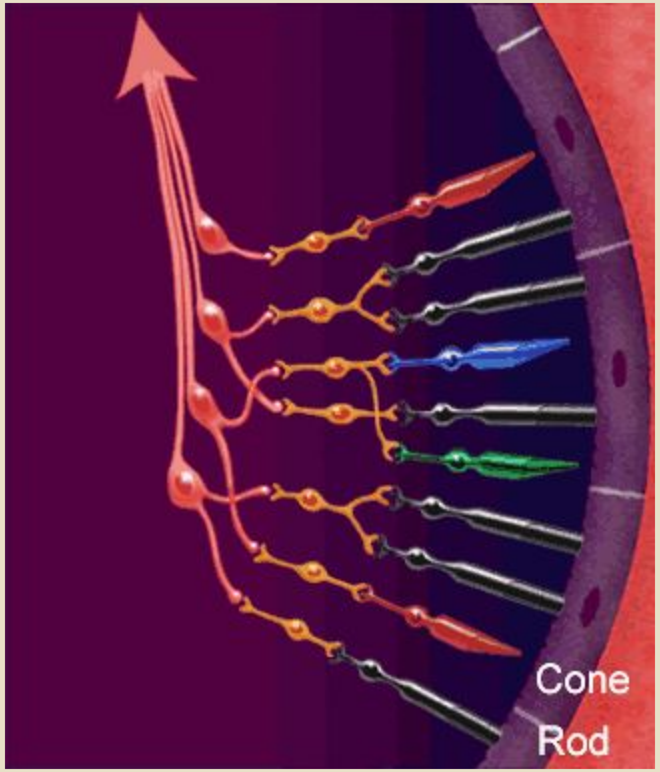
Phototransduction

- Bipolar, horizontal and amacrine cells form neuronal network; receive input from photoreceptors and process visual information.
- There are two types of glutamate receptors on bipolar cells which determine whether cell is excited or inhibited
- Metabotropic: receptors present on On-center off-surround cells. Hyperpolarization of photoreceptor decreases release of glutamate and depolarize them due to decrease inhibition. Stimulate On-ganglion cells and their axons carry impulses in optic nerve
- Ionotropic: present on Off-center, On-surround cells. Decreased glutamate release hyperpolarize these bipolar cells which in turn inhibit off-center ganglion cells and decrease their firing rate.



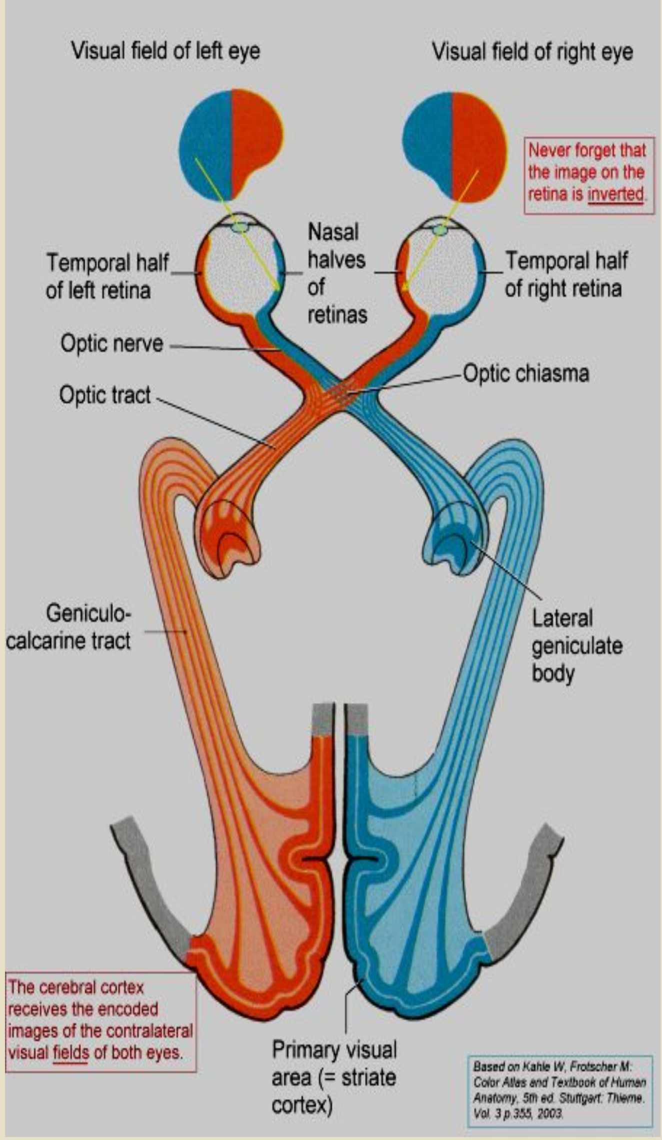
Retinal processing

- Horizontal cells are attached to surround of bipolar cells; they receive input and inhibit Bipolar cells in the area lateral to excited rods & cones to ↑ the contrast at the edge of an image (lateral inhibition)
- Amacrine cells synapse on Ganglion cells, separate movements of object from surrounding
- Fovea has only cones in a ratio of 1:1 ratio with ganglion cells; the visual acuity is maximum in this area
- Ganglion cells integrate retinal processing regarding color, brightness, contrast and pass them in optic nerve



Visual pathway

- Two optic nerves pass backwards to optic chiasma where nasal fibers cross, form optic tract consisting of nasal fibers of opposite and temporal fibers of same side.
- Few fibers from optic tract go to midbrain Pretectal nuclei (pupillary light reflex); few to Superior colliculi (coordinate head & eye movement) & few fibers to Suprachiasmatic nucleus of hypothalamus for circadian rhythm (light/ dark cycle); Also, to Reticular nuclei to alert cerebral cortex.
- Majority of fibers synapse in lateral geniculate nucleus; that separates color, form, depth, and movement
- Axons of thalamic nuclei form optic radiations and project to Primary visual area 17 in visual cortex in occipital lobe.



Processing in Visual Cortex (Area 17)

- 3 types of cells in visual cortex

  1. Simple: detect bar of light with correct position and orientation
  2. Complex: detect moving bars, lines and edges with correct orientation
  3. Hypercomplex: respond best to lines with particular length, and curves and angles (complex shapes)

- Visual cortex have topographical representation of retina with fovea having greatest representation